

Cost Analysis: Cloud vs Local Processing

Date: 2026-01-28

Hardware Arrival: February 2, 2026

Decision Point: Cloud (Dataflow) vs Local (4 Mac Studios, 1.2TB RAM total)

Executive Summary

Main Sources (vast majority of messages):

- claude_code
- claude_web
- gemini_web

Key Constraints:

- GPU enrichments **must** run locally (no GPU access on Google Cloud)
- All user messages checked by Gemini Flash Lite (spelling correction)
- Assistant messages skip spelling correction (no mistakes)

Recommendation: Wait for local hardware (Feb 2) - GPU enrichments require local anyway, and with 1.2TB RAM you can process everything in 2-3 days.

Cost Breakdown

1. Gemini Flash Lite API Costs

Pricing (as of 2026):

- Input: **\$0.100 per million tokens**
- Output: **\$0.400 per million tokens**
- Free tier: 1,000 requests/day

Assumptions:

- All user messages checked (to determine if spelling correction needed)
- Average user message: ~500 tokens (input)
- Average response: ~100 tokens (output JSON)
- ~24% of messages are user messages (from Stage 0 assessment)

Cost Calculation:

From Stage 0 assessment:

- claude_code: ~80,813 messages total
- User messages: ~19,697 (24.4%)
- Average message length: ~500 tokens

For claude_code:

- User messages to check: 19,697
- Input tokens: $19,697 \times 500 = 9,848,500$ tokens = 9.85M tokens
- Output tokens: $19,697 \times 100 = 1,969,700$ tokens = 1.97M tokens
- Cost: $(9.85 \times \$0.100) + (1.97 \times \$0.400) = \$0.985 + \$0.788 = \mathbf{\$1.77}$

For all 3 main sources (assuming similar volumes):

- Estimated total user messages: ~60,000 (3× claude_code estimate)
- Input tokens: $60,000 \times 500 = 30\text{M}$ tokens
- Output tokens: $60,000 \times 100 = 6\text{M}$ tokens
- Cost: $(30 \times \$0.100) + (6 \times \$0.400) = \$3.00 + \$2.40 = \mathbf{\$5.40}$

Gemini Flash Lite Total: ~\$5-10 (very affordable)

2. Google Cloud Dataflow Costs

Pricing Model:

- Billed per second of resource usage
- Worker vCPU: ~\$0.055/hour (us-central1)
- Worker memory: ~\$0.0045/GB/hour
- Dataflow Shuffle: ~\$0.05/GB processed

Assumptions:

- 10 workers (max_num_workers=10)
- Standard n1-standard-4 (4 vCPU, 15GB RAM per worker)
- Processing time: ~2-4 hours for 60K messages
- Shuffle data: ~10GB

Cost Calculation:**Compute:**

- $10 \text{ workers} \times 4 \text{ vCPU} \times \$0.055/\text{hour} \times 3 \text{ hours} = \6.60
- $10 \text{ workers} \times 15\text{GB} \times \$0.0045/\text{GB}/\text{hour} \times 3 \text{ hours} = \2.03
- **Subtotal:** \$8.63

Shuffle:

- $10\text{GB} \times \$0.05/\text{GB} = \0.50

Total Dataflow Cost: ~\$9-15 (one-time processing)

3. Gemini 001 Embedding Costs (Cloud)

Pricing:

- **\$0.15 per million input tokens**

- No output token cost (embeddings are output)

Assumptions:

- ~60K entities (L4, L5, L8) need embeddings
- 6 task-specific embeddings per entity
- Average text length: ~500 tokens per entity
- Total: 60K × 6 × 500 = 180M tokens

Cost: 180M tokens × \$0.15/1M = **\$27**

Note: This is only if you generate cloud embeddings. Local embeddings (Scout/Ollama, 1024-dim) are free.

4. BigQuery Storage Costs

Pricing:

- Active storage: \$0.020/GB/month
- Long-term storage (>90 days): \$0.010/GB/month

Assumptions:

- ~60K messages → ~500MB in entity_unified
- Enrichments: ~200MB
- Embeddings: ~2GB (3072-dim vectors)

Monthly Cost: ~\$0.05 (negligible)

5. Total Cloud Cost Estimate

Component	Cost	Notes
Gemini Flash Lite	\$5-10	All user messages checked
Dataflow	\$9-15	One-time processing
Gemini 001 Embeddings	\$27	Optional (can use local)
BigQuery Storage	\$0.05/month	Ongoing
Total (One-Time, with embeddings)	\$41-52	Includes cloud embeddings
Total (One-Time, no embeddings)	\$14-25	Use local embeddings instead

Local Processing (4 Mac Studios)

Hardware Specs

- **4 Mac Studios** arriving Feb 2
- **1.2TB total RAM** (300GB per Mac Studio)
- **Decent compute** (M-series chips)

- **GPUs available** (for enrichments)

Local Processing Advantages

Cost: \$0 (hardware already purchased)

Capabilities:

-  GPU enrichments (GoEmotions, KeyBERT, BERTopic, RoBERTa)
-  Can't do GPU enrichments in cloud anyway
-  1.2TB RAM can handle entire dataset in memory
-  Process in 2-3 days with parallel processing

Processing Strategy:

1. **Spelling Correction:** Gemini Flash Lite API (same cost: \$5-10)
2. **spaCy Processing:** Local (free)
3. **THE GATE:** Local (free)
4. **GPU Enrichments:** Local (free, requires GPUs)
5. **Cloud Embeddings:** Still use Gemini 001 API (3072-dim) → BigQuery
6. **Local Embeddings:** Scout/Ollama (1024-dim) → Local storage

Local Processing Timeline

With 4 Mac Studios (parallel processing):

- Day 1: Process claude_code (largest source)
- Day 2: Process claude_web + gemini_web
- Day 3: GPU enrichments + embeddings

Total: 2-3 days to process everything

Cost Comparison

Approach	One-Time Cost	Ongoing Cost	Timeline	GPU Enrichments	Cloud Embeddings
Cloud (Dataflow)	\$14-25 (+ \$27 for embeddings)	\$0.05/month	2-4 hours	 Not possible	 Yes (\$27)
Local (Mac Studios)	\$5-10 (Gemini Flash Lite) + \$27 (optional cloud embeddings)	\$0	2-3 days	 Yes	Optional

Recommendation

Wait for Local Hardware (Feb 2)

Reasons:

1. GPU Enrichments Required:

- GoEmotions, KeyBERT, BERTopic, RoBERTa need GPUs
- Google Cloud won't give GPU access
- Must run locally anyway

2. Cost Difference is Minimal:

- Cloud: \$14-25 one-time
- Local: \$5-10 (Gemini API only)
- Savings: ~\$10-15 (not significant)

3. Hardware Arrives Soon:

- Feb 2 is only 5 days away
- 1.2TB RAM can process everything easily
- 2-3 days processing time is acceptable

4. Better Control:

- Full control over processing
- Can iterate and debug locally
- No cloud vendor lock-in

5. Complete Pipeline:

- Can do ALL processing locally (except cloud embeddings)
- GPU enrichments + local embeddings in one place
- Simpler architecture

Hybrid Approach (Best of Both)

Recommended: Process locally, but use cloud for:

1. **Gemini Flash Lite API:** \$5-10 (spelling correction - all user messages)
2. **Gemini 001 Embeddings** (optional): \$27 (3072-dim cloud embeddings → BigQuery)
3. **BigQuery Storage:** Final destination for entity_unified, enrichments, embeddings

Local Processing:

- Extract, THE GATE, spaCy
- GPU enrichments (GoEmotions, KeyBERT, BERTopic, RoBERTa) - **REQUIRED locally**
- CPU enrichments (TextBlob, TextStat, NRCLx)
- Local embeddings (Scout/Ollama, 1024-dim) - **FREE**

Cloud Services:

- Gemini Flash Lite API (spelling)
- Gemini 001 API (optional cloud embeddings)
- BigQuery (storage)

Total Cost: \$5-10 (spelling only) or **\$32-37** (spelling + cloud embeddings)

Processing Plan (After Feb 2)

Day 1: Setup & claude_code

1. Set up 4 Mac Studios
2. Install dependencies (spaCy, GPU libraries)
3. Process claude_code (largest source)
 - Extract → THE GATE → Gemini Flash Lite → spaCy → L4/L5/L8
 - Write to entity_unified

Day 2: claude_web + gemini_web

1. Process claude_web
2. Process gemini_web
3. Verify all data in entity_unified

Day 3: Enrichments & Embeddings

1. **GPU Enrichments** (local):
 - GoEmotions
 - KeyBERT
 - BERTopic
 - RoBERTa Hate
 2. **CPU Enrichments** (local):
 - TextBlob
 - TextStat
 - NRCLx
 3. **Cloud Embeddings:**
 - Gemini 001 (3072-dim) → entity_embeddings
 4. **Local Embeddings:**
 - Scout/Ollama (1024-dim) → Local storage
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Cost Summary

Cloud Approach:

- Dataflow: \$9-15
- Gemini Flash Lite: \$5-10
- **Total:** \$14-25
- **Timeline:** 2-4 hours
- **GPU Enrichments:** ❌ Not possible

Local Approach:

- Gemini Flash Lite API: \$5-10 (spelling correction - all user messages)
- Gemini 001 Embeddings: ~\$15-30 (60K entities × 6 task types × ~500 tokens avg × \$0.15/1M tokens)

- **Total:** \$20-40
- **Timeline:** 2-3 days
- **GPU Enrichments:**  Yes (required - no cloud GPU access)

Wait for Local: Recommended - GPU enrichments require local anyway, and cost difference is minimal (\$14-25 cloud vs \$20-40 local, but local includes embeddings).

Next Steps

1. **Wait for hardware** (Feb 2)
 2. **Set up local processing environment**
 3. **Process main 3 sources** (claude_code, claude_web, gemini_web)
 4. **Run GPU enrichments locally**
 5. **Generate cloud embeddings** (Gemini 001 → BigQuery)
 6. **Generate local embeddings** (Scout/Ollama → Local)
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Bottom line: All costs are small (\$10–\$50 range). Nothing in the hundreds. Safe to run cloud in parallel with local—Dataflow for speed, Mac Studios for GPU enrichments and experiments.